ScikitLearn

LAB(6)

Part-1

1. **Write a Python program to load the iris data from a given csv file into a dataframe and print the shape of the data, type of the data and first 3 rows.**

2. **Write a Python program using Scikit-learn to print the keys, number of rows-columns, feature names and the description of the Iris data.**

3. **Write a Python program to get the number of observations, missing values and nan values.**

4. **Write a Python program to create a 2-D array with ones on the diagonal and zeros elsewhere. Now convert the NumPy array to a SciPy sparse matrix in CSR format.**

5. **Write a Python program to view basic statistical details like percentile, mean, std etc. of iris data.**

6. **Write a Python program to get observations of each species (setosa, versicolor, virginica) from iris data.**

7. **Write a Python program to drop Id column from a given Dataframe and print the modified part. Call iris.csv to create the Dataframe.**

8. **Write a Python program to access first four cells from a given Dataframe using the index and column labels. Call iris.csv to create the Dataframe.**

9. **Write a Python program to create a graph to find relationship between the sepal length and width.**

10. **Write a Python program to create a graph to find relationship between the petal length and width.**

11. **Write a Python program to create a graph to see how the length and width of SepalLength, SepalWidth, PetalLength, PetalWidth are distributed.**

12. **Write a Python program to create a box plot (or box-and-whisker plot) which shows the distribution of quantitative data in a way that facilitates comparisons between variables or across levels of a categorical variable of iris dataset. Use seaborn.**

13. **Write a Python program using Scikit-learn to split the iris dataset into 70% train data and 30% test data. Out of total 150 records, the training set will contain 120 records and the test set contains 30 of those records. Print both datasets.**